<u>**OSSP (Operating Systems And Systems Programming) - 25CS2104E**</u>

<u>**SKILL**</u>

1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

   To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing

Code:

gayathri@Gayathri: ~

```
 GNU nano 8.7.1                              program1.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    printf("Parent process started\n");
    printf("Parent PID: %d\n", getpid());

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
    }
    else if (pid == 0)
    {
        printf("Child process created\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID of child: %d\n", getppid());

        execlp("ls", "ls", "-l", NULL);

        printf("exec failed\n");
    }
    else
    {
        printf("Parent is waiting for child...\n");
        wait(NULL);
        printf("Child process completed\n");
    }

    return 0;
}
```

Output:

```
gayathri@Gayathri: ~
gayathri@Gayathri:~$ nano program1.c
gayathri@Gayathri:~$ gcc program1.c -o program1
gayathri@Gayathri:~$ ./program1
Parent process started
Parent PID: 533
Parent is waiting for child...
Child process created
Child PID: 534
Parent PID of child: 533
total 416
-rwxr-xr-x 1 gayathri gayathri 16080 Aug 12 07:00 Childwait
-rw-r--r-- 1 gayathri gayathri   327 Aug 12 07:02 Childwait.c
-rwxr-xr-x 1 gayathri gayathri 16176 Aug  6 04:02 Copy1toOther
-rw-r--r-- 1 gayathri gayathri   534 Aug  6 04:02 Copy1toOther.c
-rwxr-xr-x 1 gayathri gayathri 15952 Aug  7 15:23 Hello
-rw-r--r-- 1 gayathri gayathri    94 Aug  7 15:22 Hello.c
-rwxr-xr-x 1 gayathri gayathri 16304 Aug  6 03:58 Linux
-rw-r--r-- 1 gayathri gayathri   562 Aug  6 04:00 Linux.c
-rwxr-xr-x 1 gayathri gayathri 16048 Aug 12 08:07 MultipleChild
-rw-r--r-- 1 gayathri gayathri   223 Aug 12 08:08 MultipleChild.c
drwxr-xr-x 2 gayathri gayathri  4096 Jul 30 04:18 OSLAB
-rw-r--r-- 1 gayathri gayathri   193 Aug 13 04:11 PCEC
-rwxr-xr-x 1 gayathri gayathri 16000 Aug  1 05:44 ParentChild
-rw-r--r-- 1 gayathri gayathri   174 Aug  1 05:49 ParentChild.c
-rwxr-xr-x 1 gayathri gayathri 16040 Aug  1 06:05 Pcpid
-rw-r--r-- 1 gayathri gayathri   221 Aug  1 06:06 Pcpid.c
-rwxr-xr-x 1 gayathri gayathri 16032 Aug  1 06:07 Pcrun
-rw-r--r-- 1 gayathri gayathri   211 Aug  1 06:07 Pcrun.c
-rwxr-xr-x 1 gayathri gayathri 16080 Aug 12 06:50 childfork
-rw-r--r-- 1 gayathri gayathri   281 Aug 12 06:51 childfork.c
-rwxr-xr-x 1 gayathri gayathri 16176 Aug  6 04:03 copyfile
-rw-r--r-- 1 gayathri gayathri   534 Aug  6 04:06 copyfile.c
-rwxr-xr-x 1 gayathri gayathri 15992 Aug 13 04:09 exclp
-rw-r--r-- 1 gayathri gayathri   179 Aug 13 04:10 exclp.c
-rw-r--r-- 1 gayathri gayathri     0 Jul 29 06:32 file1.tx
-rw-r--r-- 1 gayathri gayathri     6 Jul 29 06:35 file1.txt
-rwxr-xr-x 1 gayathri gayathri 15992 Aug  7 15:37 fork
-rw-r--r-- 1 gayathri gayathri    90 Aug  7 15:36 fork.c
-rw-r--r-- 1 gayathri gayathri    90 Aug  7 15:37 "fork.c'"
-rw-r--r-- 1 gayathri gayathri     0 Jul 29 07:03 hello.c
-rw-r--r-- 1 gayathri gayathri   425 Aug  5 08:29 linux.c
```

```
-rw------- 1 gayathri gayathri     6 Jul 29 06:34 nano.2179.save
-rw------- 1 gayathri gayathri     6 Jul 29 06:34 nano.2238.save
-rwxr-xr-x 1 gayathri gayathri 16168 Aug 13 04:43 orphan
-rw-r--r-- 1 gayathri gayathri   314 Aug 13 04:43 orphan.c
-rwxr-xr-x 1 gayathri gayathri 16080 Aug 13 04:12 pcec
-rw-r--r-- 1 gayathri gayathri   193 Aug 13 04:15 pcec.c
-rwxr-xr-x 1 gayathri gayathri 16208 Aug 13 05:12 practical3
-rw-r--r-- 1 gayathri gayathri   403 Aug 13 05:12 practical3.c
-rwxr-xr-x 1 gayathri gayathri 16128 Aug 13 05:15 practical4
-rw-r--r-- 1 gayathri gayathri   357 Aug 13 05:14 practical4.c
-rw-r--r-- 1 gayathri gayathri   403 Aug 13 05:10 practicle3.c
drwxr-xr-x 4 gayathri gayathri  4096 Aug 13 05:29 process-lab
-rwxr-xr-x 1 gayathri gayathri 16000 Aug 12 06:41 processid
-rw-r--r-- 1 gayathri gayathri   122 Aug 12 06:42 processid.c
-rwxr-xr-x 1 gayathri gayathri 16208 Aug 18 14:42 program1
-rw-r--r-- 1 gayathri gayathri   664 Aug 18 14:42 program1.c
-rwxr-xr-x 1 gayathri gayathri 16040 Aug 13 04:20 zombie
-rw-r--r-- 1 gayathri gayathri   254 Aug 13 04:20 zombie.c
Child process completed
gayathri@Gayathri:~$
```

2. To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

   To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

Code:

```
gayathri@Gayathri: ~

  GNU nano 8.7.1                                                        program2
#include <stdio.h>
#include <stdlib.h>
#include <termios.h>
#include <unistd.h>
#include <string.h>
#define BUFFER_SIZE 100
struct termios oldt;
void set_raw_mode(int enable)
{
    if (enable)
    {
        tcgetattr(STDIN_FILENO, &oldt);
        struct termios newt = oldt;
        newt.c_lflag &= ~(ECHO | ICANON);
        tcsetattr(STDIN_FILENO, TCSANOW, &newt);
    }
    else
    {
        tcsetattr(STDIN_FILENO, TCSANOW, &oldt);
    }
}
int main()
{
    char buffer[BUFFER_SIZE];
    int index;
    char ch;

    while (1)
    {
        index = 0;
        printf("myshell> ");
        fflush(stdout);
        set_raw_mode(1);

        while (1)
        {
            ch = getchar();
```

gayathri@Gayathri: ~

```
GNU nano 8.7.1
            ch = getchar();

        if (ch == '\n' || ch == '\r')
            {
                if (index > 0)
                {
                    buffer[index] = '\0';
                    printf("\n");
                    break;
                }
            }
            else if (ch == 127 || ch == '\b')
            {
                if (index > 0)
                {
                    index--;
                    buffer[index] = '\0';
                    printf("\b \b");
                    fflush(stdout);
                }
            }

           else
            {
                if (index < BUFFER_SIZE - 1)
                {
                    buffer[index++] = ch;
                    putchar(ch);
                    fflush(stdout);
                }
            }
        }

        set_raw_mode(0);
        if (strlen(buffer) == 0)
            continue;
        if (strcmp(buffer, "exit") == 0)
```

gayathri@Gayathri: ~

```
GNU nano 8.7.1                                                          program2.
            else if (ch == 127 || ch == '\b')
            {
                if (index > 0)
                {
                    index--;
                    buffer[index] = '\0';
                    printf("\b \b");
                    fflush(stdout);
                }
            }

           else
            {
                if (index < BUFFER_SIZE - 1)
                {
                    buffer[index++] = ch;
                    putchar(ch);
                    fflush(stdout);
                }
            }
        }

        set_raw_mode(0);
        if (strlen(buffer) == 0)
            continue;
        if (strcmp(buffer, "exit") == 0)
        {
            printf("Exiting shell...\n");
            break;
        }

        printf("You entered: %s\n", buffer);
    }

    return 0;
}
```

Output:

```
gayathri@Gayathri:~$ nano program2.c
gayathri@Gayathri:~$ gcc program2.c -o prrogram2
gayathri@Gayathri:~$ ./prrogram2
myshell> hello
You entered: hello
myshell> exit
Exiting shell...
gayathri@Gayathri:~$
```